Math 1232 Summer 2025
Bonus Mastery Quiz 12
Due Saturday, August 9

This mastery quiz has two topics, and if you don't already have 4/4 on M4 or 2/2 on S8, it might be a good idea to do them. Please do your best, but don't worry if you make a minor error—your goal is to demonstrate your mastery of the underlying material.

Feel free to consult your notes, but please don't discuss the actual quiz questions with other students in the course.

Remember that you are trying to demonstrate that you understand the concepts involved. For all these problems, justify your answers and explain how you reached them. Do not just write "yes" or "no" or give a single number.

Please turn in this quiz in on Blackboard by 11:59pm.

Topics on this quiz:

- Major Topic 4: Taylor Series
- Secondary Topic 8: Parameterization

Name:

Name: ____

Major Topic 4: Taylor Series

(a) Write a power series expression for $\frac{2x^3}{5-x}$ centered a 0. What is the radius of convergence?

Name:

(b) If $f(x) = \sum_{n=1}^{\infty} \frac{2^n}{n^2+n}(x-5)^n$, compute $\frac{d}{dx} f(x)$ and $\int f(x)\, dx$.

(c) Find a formula for the Taylor series for $x^2 \cos(x^3)$ centered at 0, and write down the first three non-zero terms explicitly.

Name:

Secondary Topic 8: Parameterization

(a) Find a **clockwise** parametrization for an ellipse 10 units wide and 6 units tall (thus with major radius 5 and minor radius 3), centered at the point $(5, 2)$.

(b) Find the area inside the cardioid $r = 1 + \cos(\theta)$.